Physical Chemistry of Materials

Lecture 07

Interacting Particles

Last Class

Energy and Partition Function

- Monatomic molecules

$$E = E_{\text{trans}} + E_{\text{elec}}$$

$$z = z_{\text{trans}} z_{\text{elec}}$$

- Diatomic and polyatomic molecules

$$E = E_{\text{trans}} + E_{\text{vib}} + E_{\text{rot}} + E_{\text{elec}}$$

$$z = z_{\text{trans}} z_{\text{vib}} z_{\text{rot}} z_{\text{elec}}$$

Last Class

Electronic Partition Function

At room temperature, almost all molecules are in their ground electronic states (no thermal excitation)

$$z_{\text{elec}} = \sum_{\substack{\text{electronic} \\ \text{state } s}} e^{-E_s/k_{\text{B}}T} \approx g_0 e^{-E_0/k_{\text{B}}T}$$

Last Class

Vibrational Partition Function

- Simple harmonic oscillator
- Classical treatment

$$z_{\mathrm{vib,CM}} = \frac{k_{\mathrm{B}}T}{h\nu}$$

- Quantum treatment

$$z_{\mathrm{vib,QM}} = \frac{e^{-h\nu/2k_{\mathrm{B}}T}}{1 - e^{-h\nu/k_{\mathrm{B}}T}}$$

- For $T \to \infty$ (classical limit), $z_{\mathrm{vib,QM}} \approx \frac{k_{\mathrm{B}}T}{h\nu}$

Last Class

Rotational Partition Function

- Rigid rotor
- Classical treatment

$$z_{\text{rot}} = \frac{8\pi^2 I}{\beta h^2}$$

- Quantum treatment

$$z_{\text{rot}} = \frac{8\pi^2 I}{\beta h^2}$$

Equilibrium Constant

At equilibrium,

$$-a\mu_{\mathrm{A}} - b\mu_{\mathrm{B}} + c\mu_{\mathrm{C}} + d\mu_{\mathrm{D}} = 0$$

$$\mu_i = -RT \ln\left(\frac{z_i}{N_i}\right)$$

- Law of mass action:

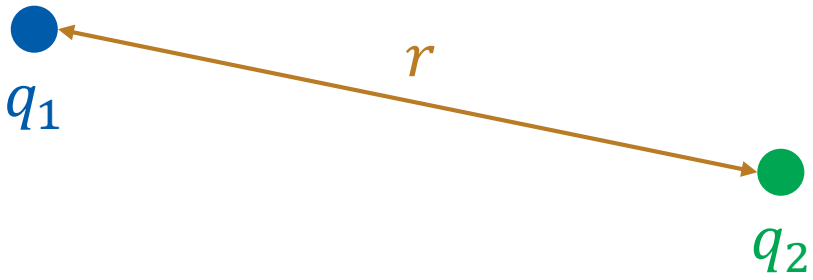
$$K_c = \frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b} = \frac{(z_{\mathrm{C}}/V)^c(z_{\mathrm{D}}/V)^d}{(z_{\mathrm{A}}/V)^a(z_{\mathrm{B}}/V)^b}$$

Intermolecular Interactions

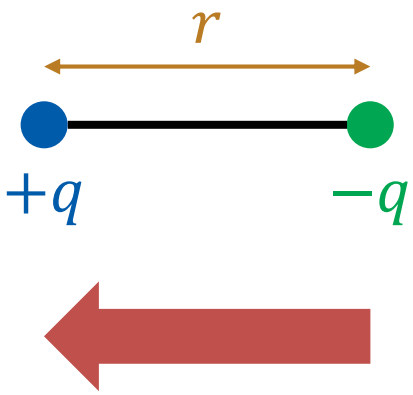
- Charge-charge interactions
- Charge-dipole interactions
- Dipole-dipole interactions
- Dispersion force
- Pauli repulsion
- Hydrogen bonds

Charge-Charge Interactions

$$V(r) = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$$



Electric Dipoles



Dipole moment

$$\vec{\mu} = q\vec{r}$$

Units: Debye

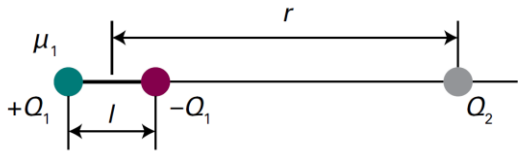
1 D = 3.34×10^{-30} C m

For polyatomic molecules,

$$\vec{\mu} = \sum_i q_i \vec{r}_i$$

Molecule	Dipole moment
CCl_4	0
H_2	0
H_2O	1.85 D
HCl	1.08 D
HI	0.42 D

Charge-Dipole Interactions



$$V(r) = \frac{1}{4\pi\varepsilon_0}\left(-\frac{Q_1Q_2}{r - l/2} + \frac{Q_1Q_2}{r + l/2}\right)$$

Letting $x = l/2r$,

$x \ll 1$ (or $l \ll r$)

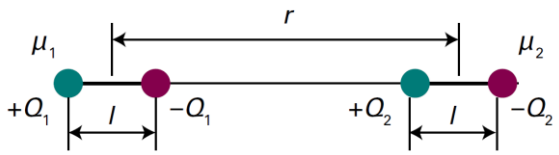
$$V(r) = \frac{Q_1Q_2}{4\pi\varepsilon_0 r}\left(-\frac{1}{1-x} + \frac{1}{1+x}\right) \approx -\frac{Q_1Q_2}{4\pi\varepsilon_0 r} \times 2x = -\frac{Q_1Q_2 l}{4\pi\varepsilon_0 r^2}$$

Hence,

$$V(r) = -\frac{\mu_1 Q_2}{4\pi\varepsilon_0 r^2}$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th Ed.)

Dipole-Dipole Interactions



$$V(r) = \frac{1}{4\pi\varepsilon_0}\left(-\frac{Q_1Q_2}{r+l} + \frac{Q_1Q_2}{r} + \frac{Q_1Q_2}{r} - \frac{Q_1Q_2}{r-l}\right)$$

Letting $x = l/r$,

$x \ll 1$ (or $l \ll r$)

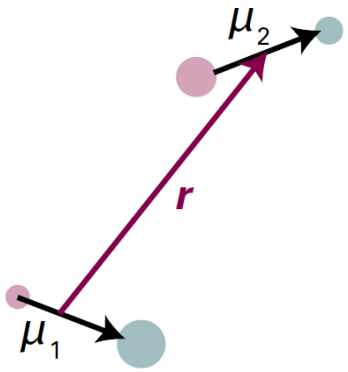
$$V(r) = -\frac{Q_1Q_2}{4\pi\varepsilon_0 r}\left(\frac{1}{1+x} - 2 + \frac{1}{1-x}\right) \approx -\frac{Q_1Q_2}{4\pi\varepsilon_0 r} \times 2x^2 = -\frac{2Q_1Q_2l^2}{4\pi\varepsilon_0 r^3}$$

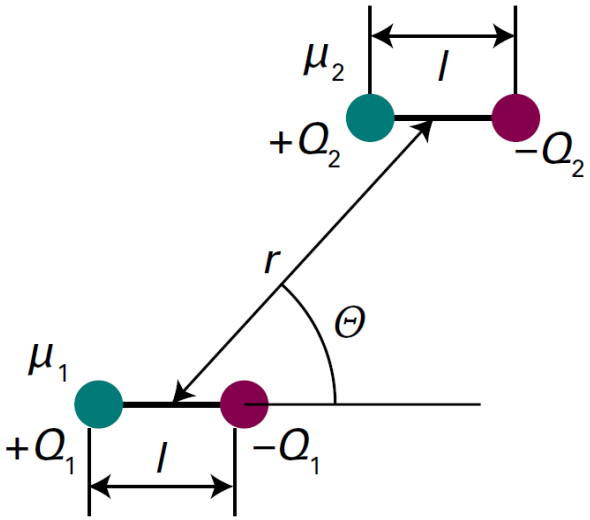
Hence,

$$V(r) = -\frac{2\mu_1\mu_2}{4\pi\varepsilon_0 r^3}$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th Ed.)

In General

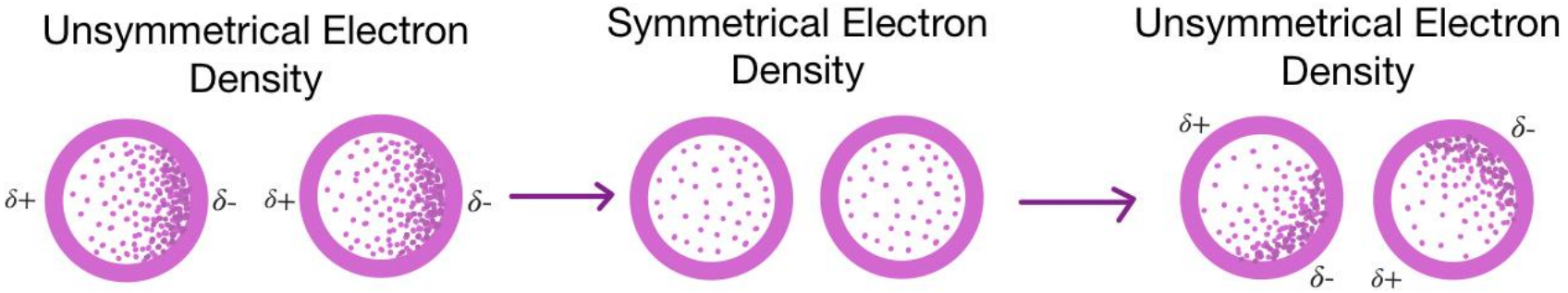




$$V(r) = \frac{\mu_1 \mu_2}{4\pi\varepsilon_0 r^3}(1 - 3\cos^2\Theta)$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th Ed.)

Dispersion Force



Induced dipole-induced dipole interaction

(Image: https://www.expii.com/t/dispersion-forces-definition-overview-8351)

Two Hydrogen Atoms

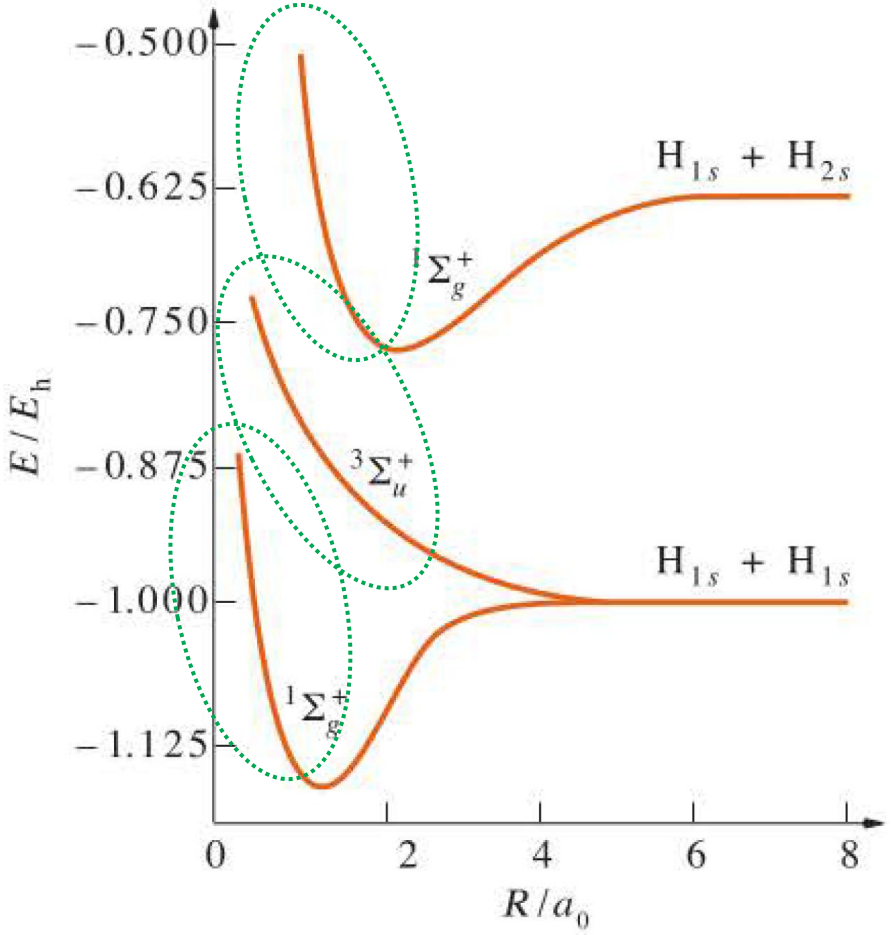


- Hamiltonian $\hat{H} = \underbrace{\hat{H}_A + \hat{H}_B}_{\hat{H}^{(0)}} + \underbrace{\hat{H}_{dip-dip}}_{\hat{H}^{(1)}}$ $\quad \hat{H}_{dip-dip} \propto \frac{\mu_1 \mu_2}{r^3}$
- Perturbation theory

$$E^{(1)} = 0$$

$$E^{(2)} \propto -\hat{H}^2_{dip-dip} = -\frac{1}{r^6}$$

Pauli Repulsion



(Image: McQuarrie, ***Quantum Chemistry***, 2nd Ed., Figure 11.12)

Approximations of Pauli Repulsion

- Hard sphere potential

$$V(r) = \begin{cases} \infty, & r < \sigma \\ 0, & r \geq \sigma \end{cases}$$

- Power-law potential (n: integer between 9 and 16)

$$V(r) = \left(\frac{\sigma}{r}\right)^n$$

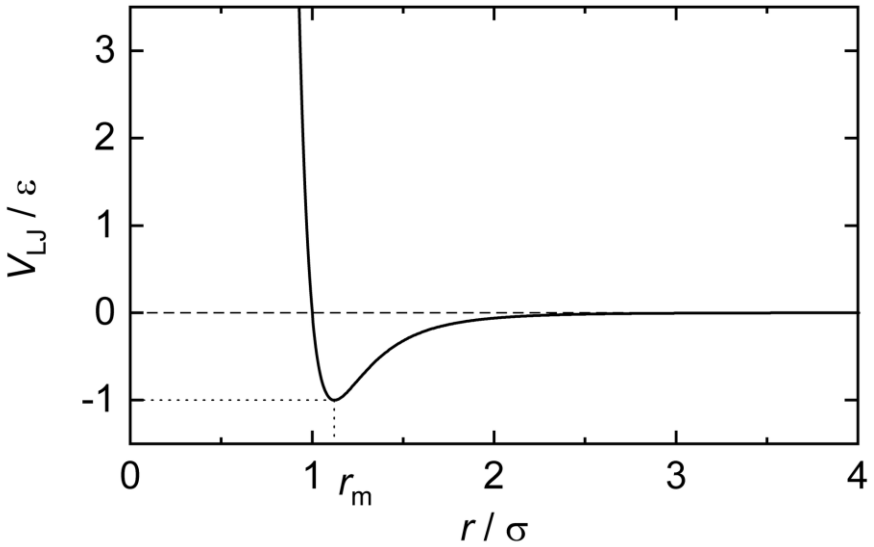
- Exponential potential

$$V(r) = ce^{-r/\sigma_0}$$

Lennard-Jones Potential

Pauli repulsion + attractive dispersion force

$$V(r) = \frac{A}{r^{12}} - \frac{B}{r^6} = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$



(Image: https://en.wikipedia.org/wiki/File:Graph_of_Lennard-Jones_potential.png)

Potential Energy Minimum

$$V(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

- Equilibrium distance r_{m}

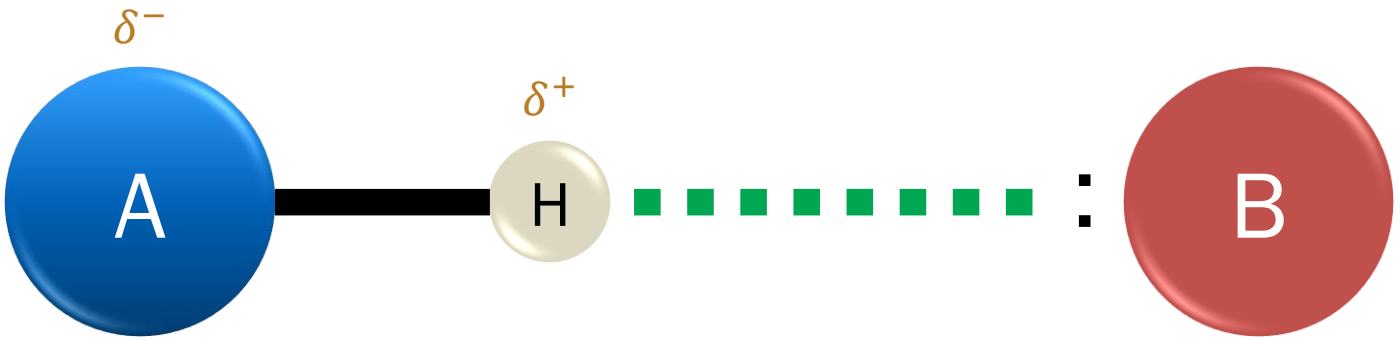
$$\frac{\mathrm{d}V}{\mathrm{d}r} = 4\varepsilon\left(-12\frac{\sigma^{12}}{r^{13}} + 6\frac{\sigma^{6}}{r^{7}}\right) = 0$$

$$r_{\mathrm{m}}^{6} = 2\sigma^{6} \text{ or } r_{\mathrm{m}} = 2^{1/6}\sigma$$

- Potential energy

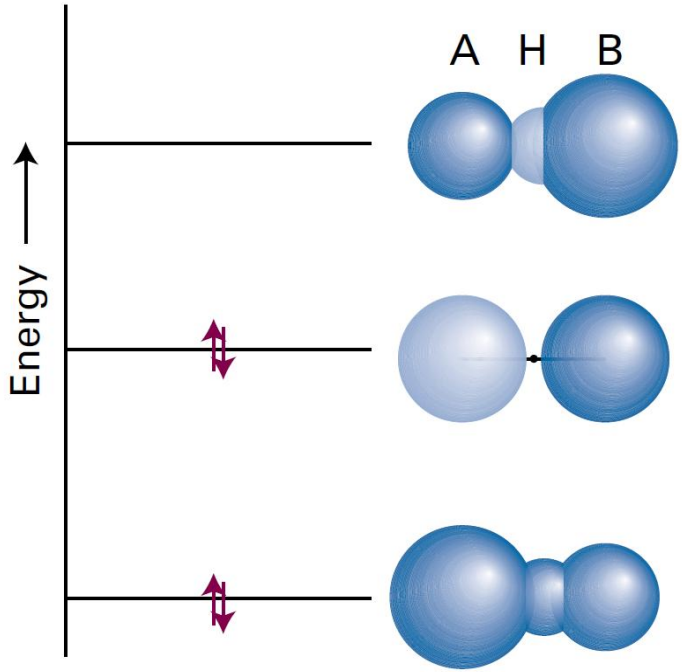
$$V(r_{\mathrm{m}}) = 4\varepsilon\left(\frac{1}{4} - \frac{1}{2}\right) = -\varepsilon$$

Hydrogen Bonds



Dipole-dipole interaction?

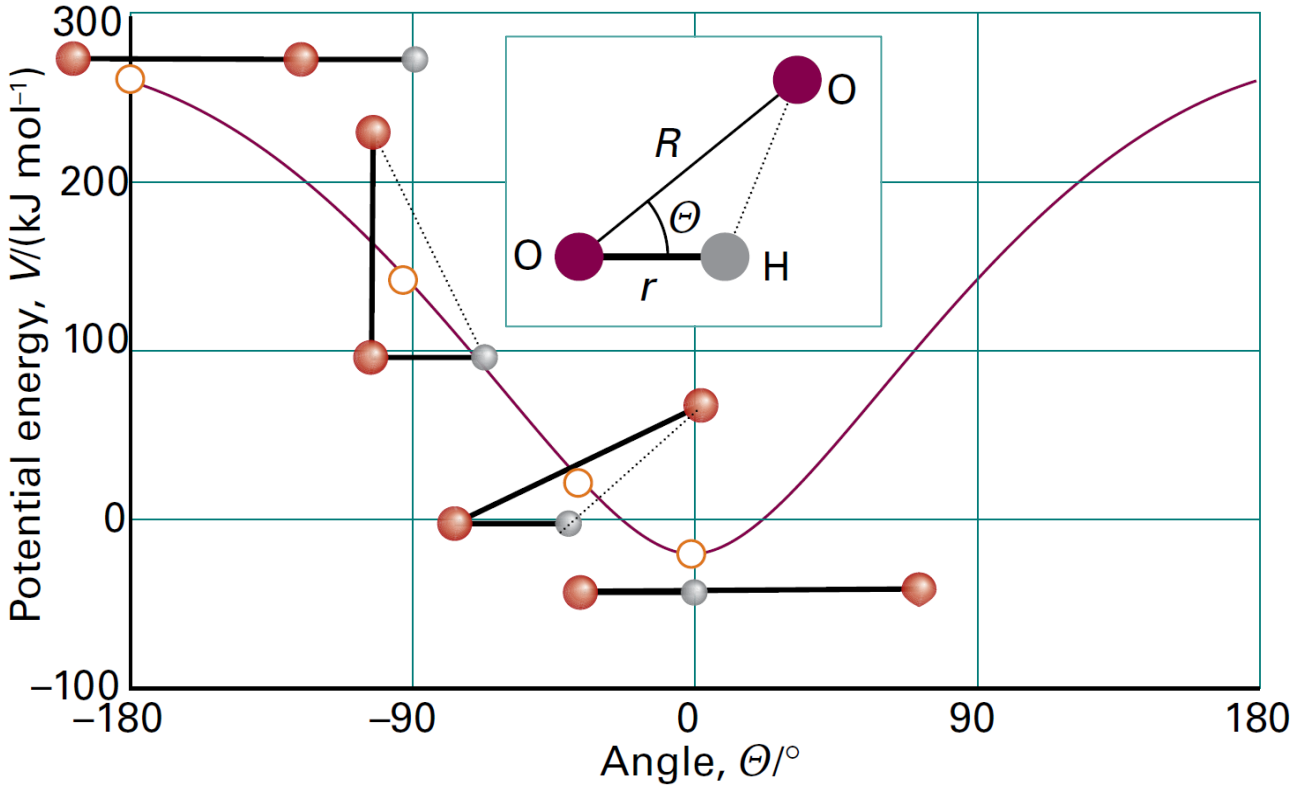
Hydrogen Bonds: MO Picture



$$\phi = c_1\psi_{\mathrm{A}} + c_2\psi_{\mathrm{H}} + c_3\psi_{\mathrm{B}}$$

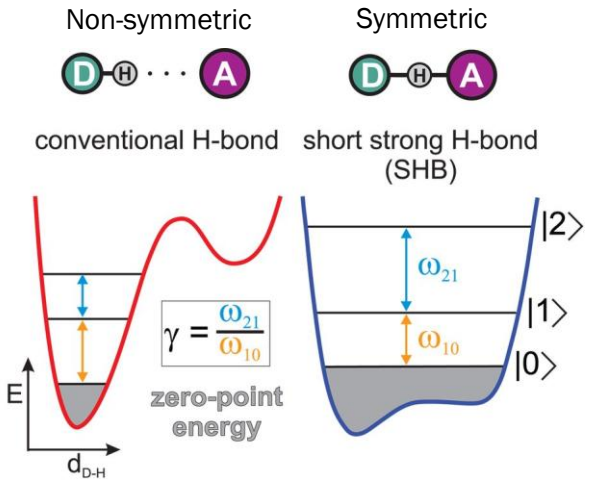
(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th Ed., Fig. 14B.5)

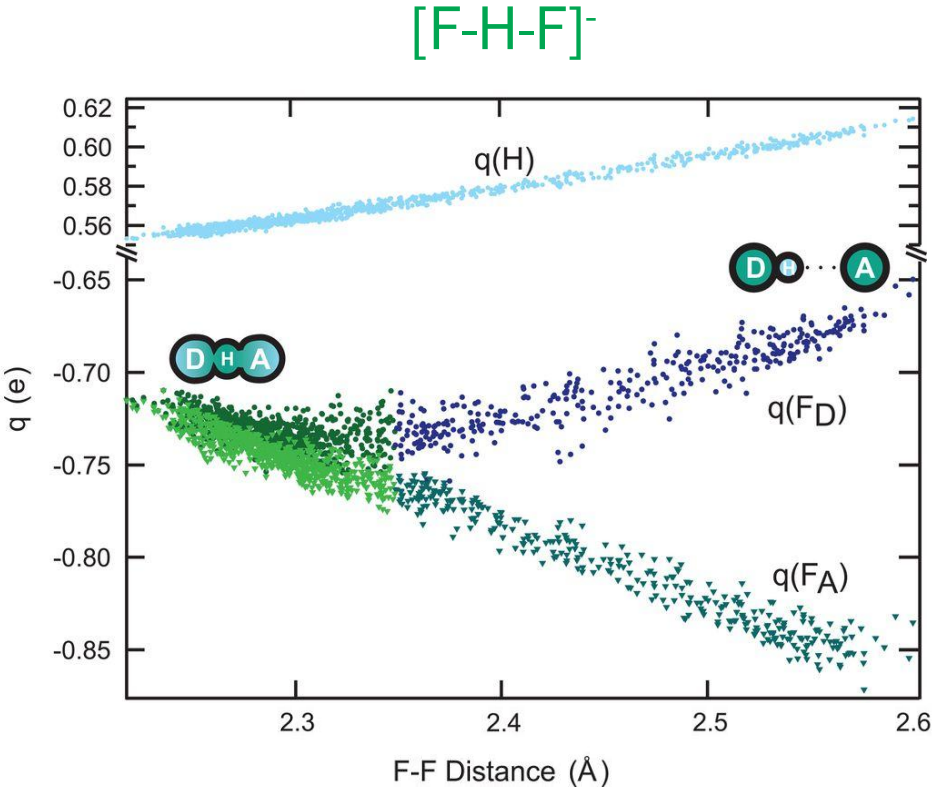
Orientation Dependence



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th Ed., Fig. 14B.6)

Covalent Bond or Dipole-Dipole?





(Image: Dereka *et al.*, ***Science***, 2021)

Summary

Interaction type	Distance dependence of potential energy	Typical energy/ ($kJ\,mol^{-1}$)	Comment
Ion–ion	$1/r$	250	Only between ions
Hydrogen bond		20	Occurs in X–H···Y, where X, Y = N, O, or F
Ion–dipole	$1/r^2$	15	
Dipole–dipole	$1/r^3$	2	Between stationary polar molecules
	$1/r^6$	0.3	Between rotating polar molecules
London (dispersion)	$1/r^6$	2	Between all types of molecules and ions

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th Ed., Table 14B.1)

Interactions and Phase Transitions

- Phase transitions: ordered ↔ disordered
- *e.g.* liquid-gas transition, magnetic transition

- Competition between energy and entropy
- Interactions are required!

(Monatomic) Gas Models

- Ideal gas
  - Energy $E = \sum_i (p_i^2/2m)$
  - Partition function

$$Z = \frac{1}{N!\,h^{3N}} \iint e^{-\sum_i p_i^2/2mk_\mathrm{B}T}\,\mathrm{d}^N\vec{r}\,\mathrm{d}^N\vec{p} = \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B}T}{h^2}\right)^{3N/2}$$

- Real gas
  - Energy $E = \sum_i (p_i^2/2m) + \sum_{i<j} E_\mathrm{pot}(r_{ij})$

Partition Function

$$E = \sum_i (p_i^2/2m) + \sum_{i<j} E_{\text{pot}}(r_{ij})$$

$$Z = \frac{1}{N!}\left(\frac{2\pi m k_{\text{B}} T}{h^2}\right)^{3N/2} \int e^{-\sum_{i<j} E_{\text{pot}}(r_{ij})/k_{\text{B}}T}\, \mathrm{d}^N \vec{r}$$

Now define

$$Z_{\text{pot}} = \int e^{-\sum_{i<j} E_{\text{pot}}(r_{ij})/k_{\text{B}}T}\, \mathrm{d}^N \vec{r}$$

Mayer f-Function

$$f_{ij} = e^{-E_{\mathrm{pot}}(r_{ij})/k_{\mathrm{B}}T} - 1$$

$$Z_{\mathrm{pot}} = \int e^{-\sum_{i<j} E_{\mathrm{pot}}(r_{ij})/k_{\mathrm{B}}T} \, \mathrm{d}^N \vec{r}$$

$$= \int \left[\prod_{i<j} e^{-E_{\mathrm{pot}}(r_{ij})/k_{\mathrm{B}}T} \right] \mathrm{d}^N \vec{r}$$

$$= \int \left[\prod_{i<j} \left(1 + f_{ij}\right) \right] \mathrm{d}^N \vec{r}$$

Mayer f-Function

$$Z_{\mathrm{pot}} = \int \left[\prod_{i<j} \left(1 + f_{ij}\right) \right] \mathrm{d}^N \vec{r}$$

$$Z_{\mathrm{pot}} = \int \left(1 + \sum_{i<j} f_{ij} + \cdots \right) \mathrm{d}^N \vec{r}$$

Assuming $f_{ij} \ll 1$ (*i.e.*, $E_{\mathrm{pot}}\left(r_{ij}\right) \ll k_{\mathrm{B}} T$),

$$Z_{\mathrm{pot}} \approx \int \left(1 + \sum_{i<j} f_{ij} \right) \mathrm{d}^N \vec{r} = V^N + \int \sum_{i<j} f_{ij} \, \mathrm{d}^N \vec{r}$$

Mayer f-Function

$$Z_{\mathrm{pot}} = V^N + \int \sum_{i<j} f_{ij}\, \mathrm{d}^N \vec{r}$$

$$Z_{\mathrm{pot}} = V^N + V^{N-1} \frac{N(N-1)}{2} \underbrace{\int \left(e^{-E_{\mathrm{pot}}(r)/k_{\mathrm{B}}T} - 1 \right) \mathrm{d}\vec{r}}_{= I}$$

$$Z_{\mathrm{pot}} \approx V^N \left(1 + \frac{N^2}{2V} I \right)$$

Helmholtz Energy and Pressure

- Helmholtz energy $A = -k_\mathrm{B} T \ln Z$
- Pressure

$$p = -\left(\frac{\partial A}{\partial V}\right)_{T,n} = k_\mathrm{B} T \left(\frac{\partial \ln Z}{\partial V}\right)_{T,n} = k_\mathrm{B} T \left(\frac{\partial \ln Z_\mathrm{pot}}{\partial V}\right)_{T,n}$$

$$= k_\mathrm{B} T \frac{\partial}{\partial V}\left[N \ln V + \ln\left(1 + \frac{N^2 I}{2V}\right)\right]$$

$$\approx k_\mathrm{B} T \left(\frac{N}{V} - \frac{N^2 I}{2V^2}\right)$$

Evaluation of I

$$I = \int \left(e^{-E_{\mathrm{pot}}(r)/k_{\mathrm{B}}T} - 1\right) \mathrm{d}\vec{r} = \int_0^{\infty} \left(e^{-E_{\mathrm{pot}}(r)/k_{\mathrm{B}}T} - 1\right) 4\pi r^2 \mathrm{d}r$$

Need to specify $E_{\mathrm{pot}}(r)$

- Sutherland potential: hard sphere + dispersion force

$$E_{\mathrm{pot}}(r) = \begin{cases} +\infty, & r < r_0 \\ -\epsilon_0 \left(\dfrac{r_0}{r}\right)^6, & r \geq r_0 \end{cases}$$

Two Regions

$$I = \int_0^{\infty} \left(e^{-E_{\mathrm{pot}}(r)/k_{\mathrm{B}}T} - 1\right) 4\pi r^2 \mathrm{d}r$$

$$E_{\mathrm{pot}}(r) = \begin{cases} +\infty, & r < r_0 \\ -\epsilon_0 \left(\frac{r_0}{r}\right)^6, & r \geq r_0 \end{cases}$$

$$I = \int_0^{r_0} \left(e^{-\infty/k_{\mathrm{B}}T} - 1\right) 4\pi r^2 \mathrm{d}r + \int_{r_0}^{\infty} \left(e^{\beta\epsilon_0 (r_0/r)^6} - 1\right) 4\pi r^2 \mathrm{d}r$$

Final Result

$$I = -\frac{4}{3}\pi r_0^3 + \int_{r_0}^{\infty} \left(e^{\frac{\epsilon_0}{k_B T}\left(\frac{r_0}{r}\right)^6} - 1 \right) 4\pi r^2 \mathrm{d}r$$

Assuming $\epsilon_0 \ll k_B T$,

$$e^{\frac{\epsilon_0}{k_B T}\left(\frac{r_0}{r}\right)^6} \approx 1 + \frac{\epsilon_0}{k_B T}\left(\frac{r_0}{r}\right)^6$$

$$I = -\frac{4}{3}\pi r_0^3 + \int_{r_0}^{\infty} \frac{\epsilon_0}{k_B T}\left(\frac{r_0}{r}\right)^6 4\pi r^2 \mathrm{d}r = -\frac{4}{3}\pi r_0^3 \left(1 - \frac{\epsilon_0}{k_B T}\right)$$

Pressure

$$p = k_{\mathrm{B}}T\left(\frac{N}{V} - \frac{N^2 I}{2V^2}\right), \qquad I = -\frac{4}{3}\pi r_0^3\left(1 - \frac{\epsilon_0}{k_{\mathrm{B}}T}\right)$$

Substitution leads to

$$p = \frac{Nk_BT}{V} + \frac{N^2 k_{\mathrm{B}}T}{V^2}\cdot\frac{2}{3}\pi r_0^3 - \frac{N^2}{V^2}\cdot\frac{2}{3}\pi r_0^3\epsilon_0$$

$$p + \frac{N^2}{V^2}\cdot\underbrace{\frac{2}{3}\pi r_0^3\epsilon_0}_{a} = \frac{Nk_{\mathrm{B}}T}{V}\left(1 + \frac{N}{V}\cdot\underbrace{\frac{2}{3}\pi r_0^3}_{b}\right)$$

Equation of State

$$p + a\frac{N^2}{V^2} = \frac{Nk_\mathrm{B}T}{V}\left(1 + b\frac{N}{V}\right)$$

As $Nb/V \ll 1$,

$$\frac{Nk_\mathrm{B}T}{V} = \frac{p + aN^2/V^2}{1 + bN/V} \approx \left(p + \frac{aN^2}{V^2}\right)\left(1 - \frac{bN}{V}\right)$$

$$\left(p + \frac{a}{V_\mathrm{m}^2}\right)(V_\mathrm{m} - b) = RT$$

(van der Waals equation of state)